

This is the initial content for java.pdf.

Java is a high-level, class-based, object-oriented programming language that is designed to have as fe

Key Features of Java:

1. Platform Independent: Java code runs on any device that has the JVM installed.
2. Object-Oriented: Everything in Java is treated as an object which helps in modular programming and
3. Simple and Familiar: Java is designed to be easy to learn and use, especially for programmers famili
4. Secure: Java provides a secure environment for executing code by confining program execution thro
5. Robust: Java emphasizes early checking for possible errors, runtime checking, and eliminates error-
6. Multithreaded: Java supports multithreading, which allows concurrent execution of two or more threa
7. Distributed: Java is designed for the distributed environment of the internet, enabling programs to sh

Basic Syntax:

- Java programs start execution with the main() method.
- Case-sensitive language where keywords and identifiers matter.
- Statements end with a semicolon.

Example:

```
```java
public class HelloWorld {
 public static void main(String[] args) {
 System.out.println("Hello, World!");
 }
}
```
```

Java Ecosystem:

- Java Development Kit (JDK): The full-featured SDK for Java development.
- Java Runtime Environment (JRE): Provides libraries and JVM to run Java applications.
- Integrated Development Environments (IDEs) like Eclipse, IntelliJ IDEA, and NetBeans.

Common Java APIs:

- Java Collections Framework
- Java IO
- Java Networking

Java is widely used for building enterprise-scale web applications, Android apps, and large systems du